

Final Project (for those not using their own data)

Due May 29 at 10 p.m. (May 27 for graduating students)

Sports are a common extra-curricular activity for young people in the United States. There is a longstanding association between participating in organized sports and academic achievement. However, the causal effect remains unclear. Advocates argue that a healthy body promotes healthy cognitive development, and that sports teach resilience, leadership, and discipline that can also promote academic learning behaviors. Skeptics point out that participation in those activities is disproportionately available to children of more advantaged families, and many school-organized sports programs require minimum grade averages to participate, increasing selection bias.

Your task is to investigate the link between participating in organized sports and reading achievement in the first grade. You will use data from the Early Childhood Longitudinal Survey – Kindergarten cohort class of 2010-11 (the files are called *eslsk_1.dta* and *eclsk_2.dta*). I've cleaned the data and provided you with a limited set of covariates. The outcome variable is READ1, a reading test score at the end of 1st grade. The treatment variable is SPORTS, a binary variable indicating whether or not the child participates in organized sports activities. Child-level control variables include gender, racial minority status, a composite measure of SES, a binary variable indicated whether the child participated in organized artistic activities, a reading score measured at the end of Kindergarten, and age in months. At the school level, you have data on school sector (coded Catholic and Other Private, with public schools as the reference group) and the school's kindergarten enrollment (a measure of school size).

You will tackle this question using two methods: covariance adjustment and propensity score stratification. As before you should produce a write up that includes the models you run and selected output. You may consult with each other for this project, but every student is expected to conduct their own analysis and produce their own write up independently. You must also submit the .html outputs for your propensity score model and final analytic models on Canvas, separately from your write up (which must be in Word format).

I. Covariance Adjustment

A. Consider a naïve model controlling for no covariates

Include only the treatment variable (sports participation), and allow its slope to vary. Interpret the intercept, coefficient, and random effects. Explain in plain language what you found.

B. Controlling for relevant covariates

Include all covariates at both levels, with fixed slopes. Interpret the coefficient for your treatment effect. Explain in plain language why your treatment effect estimate changed, as well as what else you learned from your model.

II. Propensity Score Stratification

The aim here is to estimate the association between sports participation and reading achievement within subsets of students who have the same, or nearly the same, probability of exposure to sports participation

A. Building a Propensity Model. Using the pre-treatment covariates available in dataset, use logistic regression to predict the treatment (sports participation). Use a two-level logistic regression model. Make sure to include all covariates that contribute to the prediction of exposure or the outcome. Save the predicted log odds of treatment.

B. Ensure Common Support. Identify the region of common support and remove those outside of it from your analytic sample. Explain why this is necessary.

C. Stratification. Sort the students by their propensity scores and subdivide them into 5 equal-sized strata.

D. Check Balance. Check the balance between treated and control cases on the propensity score and all covariates using standardized mean differences. [Hint: You can do this manually, or create a function to expedite the work.] What proportion of the within-strata differences have SMDs below .2? Below .1? Describe the quality of the balance generally, and what this will mean for potential bias in your estimates.

E. Study the association between treatment and the outcome. Run your final analytic model in HLM. Interpret the treatment effect. Explain in plain language what you found.

E. Consider whether your estimated effect can be interpreted causally.

Explain the assumptions that would be required for your estimate of the effect of sports participation to be interpreted causally. Provide an assessment of the extent to which these assumptions are believable.

F. Sensitivity Analysis. Conduct a sensitivity analysis to evaluate how robust your estimated effect is to potential unobserved confounding variables and explain what you found.

III. Conclusion

Write a few summary paragraphs explaining what you learned about the association between sports participation and reading achievement in substantive terms. Compare the findings the two approaches above, and the advantages of each method.